7.4 → Trending

→ Methods for determining trends

→ i) simple average of percent changes → Not can be for premium trends that are stable

→ ii) Linear & exponential trend fitting → ——— Loss trends - - - volatile

→ (1 + pure premium trend) = (1 + frequency trend) * (1 + severity trend)

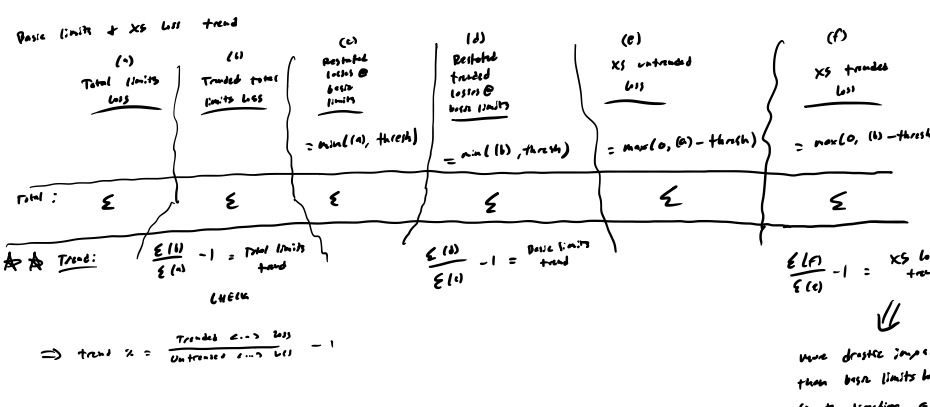
⇒ if estimating PP trend, cut off claims like usual, then divide by exposures at end
⇒ calculate Δ % to find trend

→ Effects of limits on trend

→ In general, for a positive severity trend,
Basic Limits trend ≤ Total Limits trend ≤ Excess Loss trend

→ on the other hand, if the severity trend is negative it relationship becomes
Excess loss trend ≤ Total limits trend ≤ Basic limits trend

→ Basic limit & XS loss trend



⇒ trend % = (Trended ... loss / Untrended ... loss) - 1

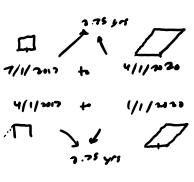
⇓
more drastic impact than basic limits loss trend (in the direction of the total limits loss trend)

→ Determining trend periods!!! Premium data

↻ when forecasting EP, can use written dates OR Earned dates

⇒ Both should result in same length of trend period, just a shifted interval by half a policy period

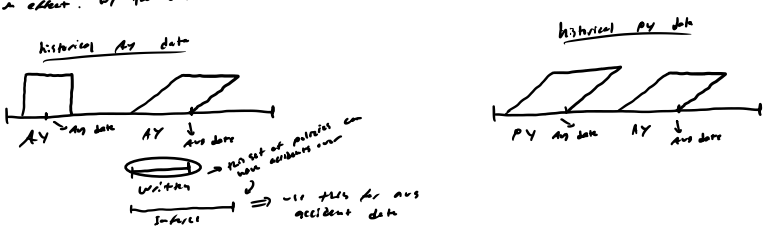
⇒ Calculate & using both methods as a CHECK!!!



→ When calculating a trend from the data, be sure to do it on an annual basis

→ Forecasting losses

→ Trend period is from any accident date in each historical period to the average accident date when the new rates will be in effect. W/ the uniform assumption, the average accident date is the midpoint of the periods.



→ Applying a trend

→ If freq &/or sev trends are clearly changing, do two-step trending ⇒ e.g. can do one-step for freq & two-step for sev

→ Two-step trending → loss data (ie w/ different fits)
① trend to avg acc date of latest historical period
② trend to avg acc date of future policy period

→ If there is a notable change in % data ⇒ ① Trend to exactly that date, then ② Trend beyond

→ Premium data → If can get Avg WP & Avg EP (i.e. $\frac{WP}{WE} \cdot \frac{EP}{EE}$) at t_1 & t_2

⇒ current trend step from t_1 to t_2 = (most recent WP at current level) / (historical EP at current level)

⇒

CY	Avg accep	Avg accep
2021	(b)	(d)
2024	(c)	

current trend factor = (c)/(b) ⇒ then use projected premium trend factor

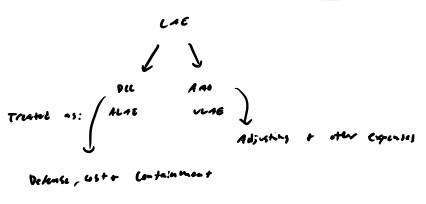
★ ↳ apply to historical EP for LR

⇒ projected trend step is just like one-step trending

7.5 → Expenses & Profit

→ LAE → ULAE, ALAE

OR



→ three methods for incorporating UW expenses:
→ 1) All variable expense method } → have historical precedence
→ 2) Premium-based projection method
→ 3) Exposure/Policy-based projection } → addresses distortions of other two, which is particularly relevant for fixed expenses make up a significant portion of total expenses

→ All variable expense method
→ all expenses are treated as variable to premium
→ Steps to calculate the UW expense provision:
→ 1) Derive the expense ratio for each year & each category → Don't use OLEP when calculating ratios ⇒ need % of historical premium
→ 2) Select a ratio for each expense type (should be representative of future)
→ 3) Sum the selections for each expense category to find the total UW expense provision

→ Potential distortions → If there are any constant (or almost constant) expenses for each risk ⇒ can result in inaccurate expense estimates (ie when not to use this approach)
→ B/c expenses vary by premium ⇒ inequity: undercharge risks w/ below-average premium & overcharge risks w/ above-average premium

→ Trending → Even though UW expenses likely increase w/ inflation, b/c they are expressed as a % of premium, this is taken into account ⇒ no trending needed

→ under all-variable expense method, VPLR = PLR = 1 - VER - Profit

→ The other two methods → 1) Premium-based projection → Fixed = (total expenses * % of fixed expenses) / WP or EP
Variable = (total expenses * % of variable expense) / WP or EP

→ 2) for exposure-based projection → Fixed = (total expenses * % of fixed expense) / WE or EE (EE = exposure)
Variable: same

→ Summary of trending expenses
Fixed → premium-based projection → only IF fixed expenses trend different than premium
→ Exposure-based projection → Yes if exposure base is not inflation-sensitive; Yes if fixed expense trends different than exposure
Variable → never

→ Additional details
→ Premium-based projection method → variable & fixed expenses are handled separately
→ important for companies w/ significant amounts of each
→ the steps for determining the UW expense provision are:
→ If the average fixed expense per exposure is needed instead, this can be found as
⇒ Fixed expense per exposure = Fixed expense ratio * Projected average Premium
→ Distortion → the fixed expense ratio will be distorted if historical & projected premium levels differ materially

→ Exposure/Policy-based method → The historical expense ratios for each category use the following data:

Expense	Data Used	Fixed Divided By	Variable Divided By
General Expense	Countrywide	Earned Exposure	Earned Premium
Other Acquisition	Countrywide	Written Exposure	Written Premium
Commissions and Brokerage	Countrywide/State	Written Exposure	Written Premium
Taxes, Licenses, and Fees	State	Written Exposure	Written Premium

→ This procedure results in the average fixed expense per exposure. So if the projected fixed expense ratio is needed instead, it can be found as

→ Projected Fixed Expense ratio = (Avg projected fixed expense per exposure) / (projected Avg Premium)

→ Potential distortions (which can be remedied slightly (not written here though))
→ While the Exposure/policy-based projection method corrects the distortions caused by the other two methods, it still has its own shortcomings. These are discussed below, along w/ potential enhancements that may be needed
→ 1) Expense categorization into fixed & variable components often relies on judgement.
→ 2) UW fixed expenses are allocated to states based on exposure or policy distribution, but actual average fixed expense levels may differ by location (e.g. higher advertising costs in some states)

→ Profit
→ VPLR = 1 - Variable expense % - Target UW profit %
↓ = 1 - V - Q_T
→ Total PLR = 1 - (F + V + Q_T)
↳ Fixed expense %

→ Random → if state a concept as a solution, explain the concept as well
e.g.) Segment the market & charge different rates
⇒ How charge different rates?? ⇒ Higher premium for higher MAGE & lower premium for lower risks

→ Sometimes you just have to read the problem & choose an interpretation that makes sense. If you're worried that your assumption isn't correct, just state what your assumption is & move forward anyway. Unless your assumption is illogical, that's probably enough for full credit

7.6 → Overall rate level indications

→ pure premium method → Indicated Avg rate = (pure premium including LAE + Fixed UW expense per exposure) / (1 - Variable UW expense % - Target UW profit %)

→ P_S = $\frac{(L + E_L) + E_F}{1 - V - Q_T}$ → VPLR → Fixed expense for a pol = (projected fixed expenses) / (# policies) * per policy

→ per exposure: $\bar{P}_S$ = $\frac{(\bar{L} + \bar{E}_L) + \bar{E}_F}{1 - V - Q_T}$

→ If developing & trending losses & other adjustments
Loss + LAE = Non-cat Ult & ALAE * LDF * ULAE factor * Current trend factor * Projected trend factor
⇒ ÷ by exposures to get PP

→ Loss ratio method → Indicated avg rate change = (Loss & LAE ratio + Fixed expense ratio) / (1 - Variable UW expense % - Target UW profit %) - 1

→ Derivation → Indicated change factor = $\frac{P_S}{P_L}$ (P_S → from PP method; P_L → Premium at CRL)
↓ = $\frac{\text{Ult \& LAE ratio} + F}{1 - V - Q_T}$ (F → fixed expense ratio)

⇒ Indicated avg rate change = Indicated change factor - 1

→ When comparing loss ratios for amortizing exposures need to be accounted for
⇒ ie. OLEP, developed losses, trended losses
⇒ ensures apples to apples comparisons across years/classes/etc.

→ Selecting final LR
⇒ use straight-average if stable / assume fluctuations are just random fluctuations
⇒ use weighted-average if results are more volatile

→ Comparison of methods
→ There are three main differences between the pure premium method (PP method) & the LR method (LR method).
→ 1) → The underlying loss measure used in the calculations is different. The PP method utilizes pure premium & the LR method utilizes the loss ratio.
→ premium at CRL is needed for the LR method, but not the PP method.
→ Clearly defined expenses are needed for the PP method, but not the LR method.
→ 2) → The two methods result in different outputs. The PP method results in an indicated rate, while the LR method results in an indicated rate change.
→ Since the LR method involves calculating the percent change from current rates, this method cannot be used for a new company that wants to determine an indicated rate. In that case, external data or judgement can be used to estimate expected PP + expense provisions + to select a target profit provision to use in the PP method.

→ Converting between indicated average rate & indicated avg rate change
⇒ $\frac{\text{ind rate}}{\text{current avg rate}} - 1$ = ind rate change
(1 + ind rate change) * Avg current rate = ind avg rate

→ Results from PP method vs LR method should be equivalent (assuming consistent data & assumptions)

Stillwater →

(a) Year ended	(1) Current Level EP	(2) Cumulative trend factors	(3) Trended Current Level EP	(4) Non-cat reported Loss + LAE	(5) LDFs	(6) Ultimate non-cat Incurred Loss + LAE	(7) Loss trend factors	(8) Trended Ultimate Non-cat Incurred Loss + LAE	(9) Ultimate non-cat incurred Actual Loss + LAE ratio	(10) Ay weights
2020-Qtr	(1)	(2)	(1) * (1+2%)^time	(4)	(5)	(4) * (5)	(7)	(6) * (1+3%)^time	(8)/(3)	(10)

(11) Ay weighted partial loss + LAE ratio = Σ years (10) * (9)
(12) Credibility = (12)
(13) Complement of credibility = (12)
(14) Credibility-weighted trended Ultimate Incurred Loss + LAE ratio = (12) * (11) + (1 - (12)) * (13)
(15) Modeled cat loss = (15)
(16) Margin for riskiness = (16)
(17) Total indicated loss + LAE ratio = (14) + (15) + (16)
(18) Permissible Loss + LAE ratio = (18) → = 1 - VER
(19) Indicated rate level needed = (17)/(18) - 1

7.1 → Risk classification

→ Importance of equitable rates
→ Goal is to segment in order to charge the true rates to each group in the population, thus avoiding adverse selection.
→ Adverse selection occurs when you charge the same rate instead of segmenting
⇒ As a result, over/under charge certain groups & over time start attracting more & more risky policies
⇒ unprofitable
⇒ rate changes
↻ repeat ...
→ until either:
i) segment in a more refined manner,
2) Become insolvent, or
3) specialize in insuring high-risk insureds only
→ Opposite of this is favorable selection.
→ "skimming the cream" is using risk characteristics to identify, attract, & select the lower-risk insureds (marketing) (underwriting)

→ The criteria for evaluating the appropriateness of rating variables can be grouped into four categories:

→ 1) Statistical Criteria
→ a) Statistical significance → different levels of the rating variable should have statistically significant & stable over time different expected costs that are
→ b) Homogeneity → The risks within the same level should have similar expected costs
→ c) Credibility → Each level should be large enough to allow credible estimate of costs
↳ balancing act in terms of how fine to make the groups
→ reasons to not include a variable are just the negative or opposite of these

→ 2) Operational Criteria
→ a) Objective → clear & measurable
→ b) Inexpensive to administrate → cost to obtain shouldn't outweigh potential benefit
→ c) Verifiable → not easily manipulated & easily verifiable

→ 3) Social Criteria
→ a) Affordability → Insurance should be affordable for all risks (even high-risks) ↳ can offer low coverage
→ b) Causality → A clear cause-&-effect between a rating variable & expected losses will increase the social acceptability of the classification
→ c) Controllability → Policyholders generally prefer to have some degree of control over the class to which they belong & the ability to influence the premium charged (i.e. a home discount, whether or not a security system is installed)
→ d) Privacy Concerns → not intrusive or privacy of insured
↳ balancing act in practice

→ 4) Legal criteria → follows applicable laws

→ Classification systems → groups risks w/ similar characteristics / loss propensity → Both use historical data & risk characteristics to determine a rate
→ appropriate when putting a homogeneous group of risks

→ Individual risk rating → Tailored to the individual insured

→ Standard rates set by the state via neither rating plan

→ Wording of question ⇒ "EVALUATE" means DESCRIBE + RECOMMEND (yay/nay)